Map Spotlight 8: Map Design

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Process of Map Design

- Step 1:
  - Combine imagination and creativity to think about possible design methods, involving topics such as map purpose, format, data, and reference mapping techniques.

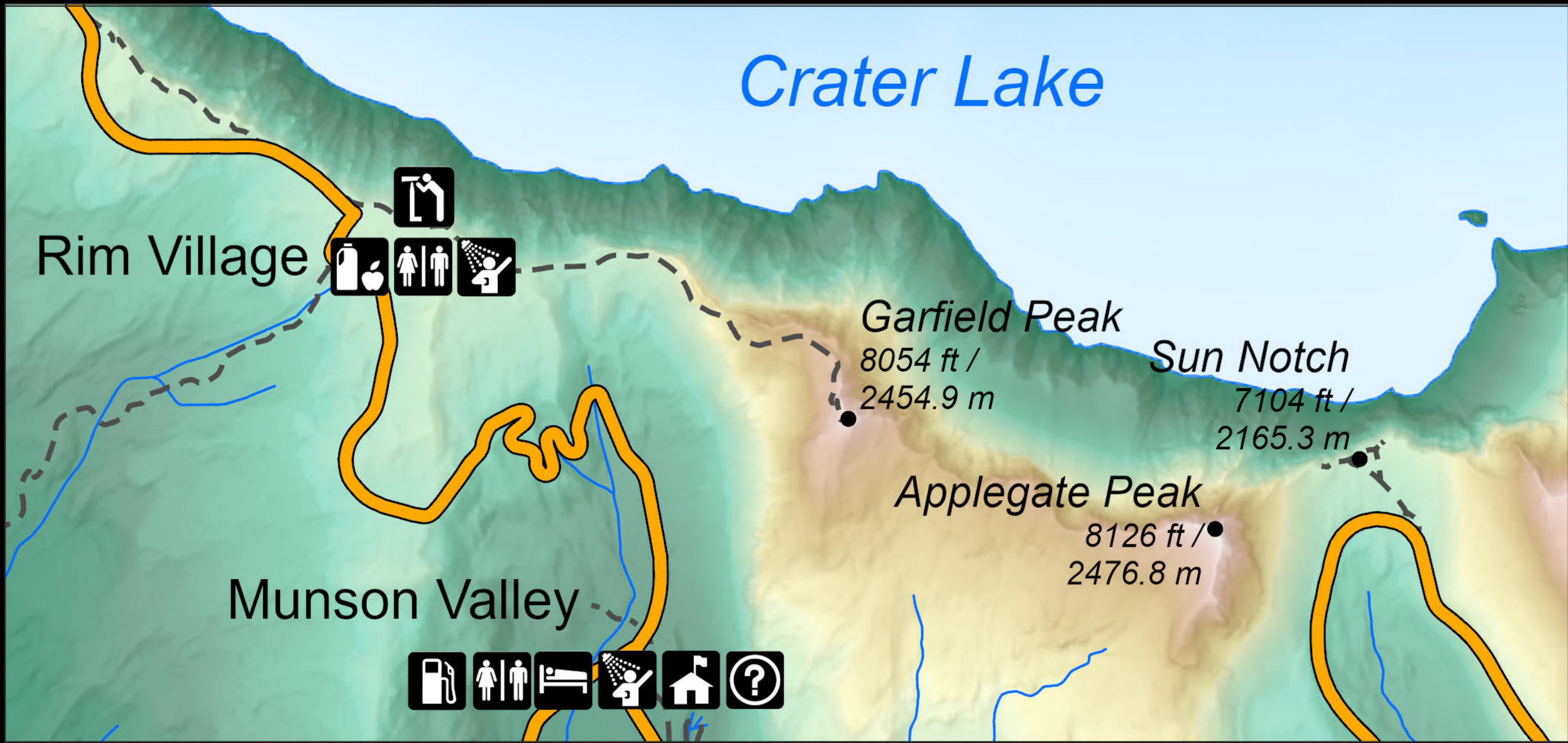
- Step 2:
  - Formulate a specific graphic plan. Consider different design strategies within the scope set in the first stage, such as the types of symbols, grading restrictions, lines, and colors, etc.

- Step 3:
  - The detailed plan of the map production includes all drawing methods and procedures, mainly the design, weight, color, and note size of the symbols.

Principles of Map Design

- Legibility
- Visual contrast
- Figure-ground organization
- Hierarchical organization

Legibility (易讀性)



Legibility (易讀性)

- Need to choose the proper graphic marks and portray them clearly.
  - The shape of point symbols must not be confusing.
  - Lines must be clearly distinguishable in form and width.
  - Differences between colors, patterns, and shadings used to differentiate between symbols must be visually distinct.
- One important aspect of legibility is size.

Recommended Minimum Symbol Sizes

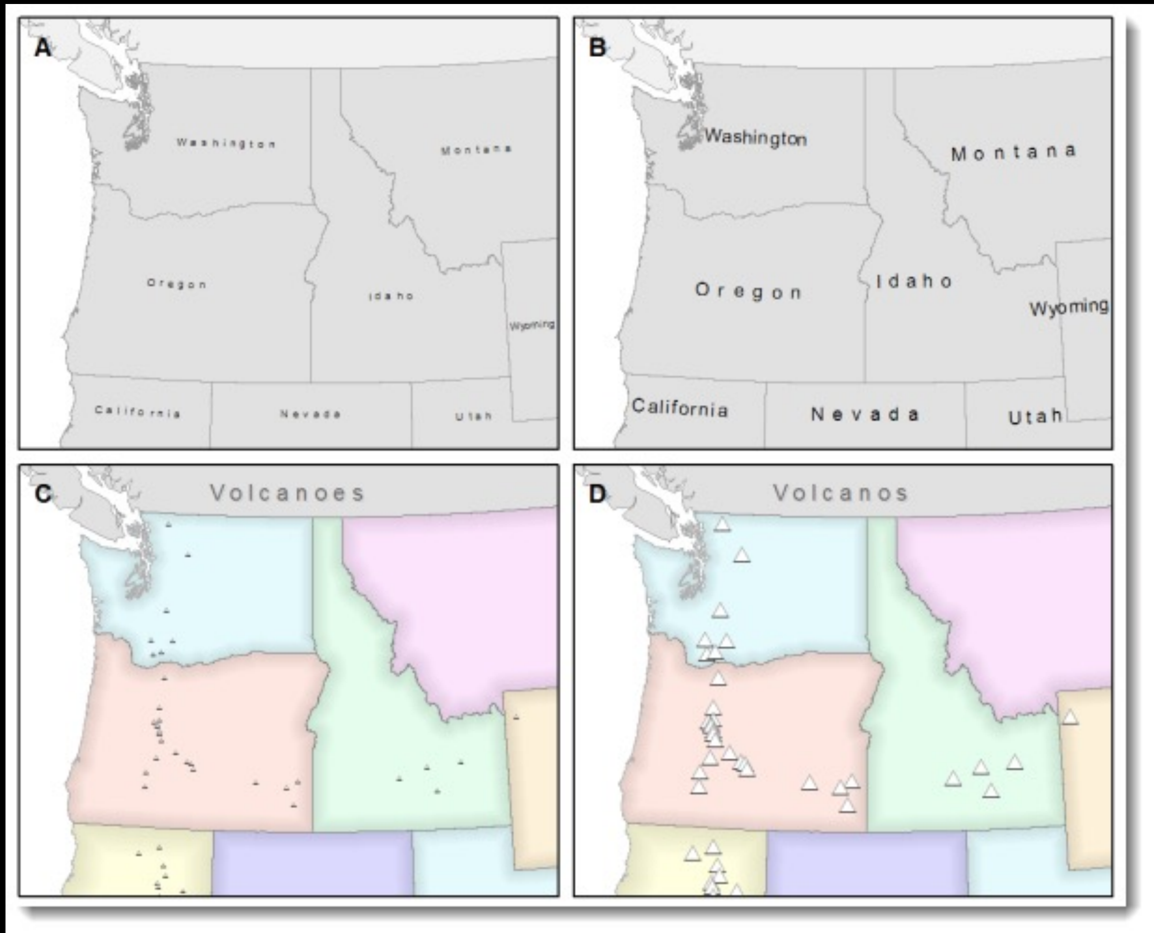
Viewing Distance (feet)	Computer Screen (points)	Printed Maps (points)	Computer Screen (inches)	Printed Maps (inches)	Computer Screen (millimeters)	Printed Maps (millimeters)
1.5	6	4	0.08	0.06	2.0	1.5
2	8	6	0.10	0.08	2.7	2.0
3	11	8	0.16	0.12	4.0	2.9
4	15	11	0.21	0.15	5.3	3.9
5	19	14	0.26	0.19	6.6	4.9
10	38	28	0.52	0.38	13.3	9.8
20	75	55	1.05	0.77	26.6	19.5
30	113	83	1.57	1.15	39.9	29.3
50	188	138	2.62	1.92	66.5	48.8
100	377	276	5.24	3.84	133.0	97.5

Recommended Minimum Text Sizes

Viewing Distance (feet)	Computer Screen (points)	Printed Maps (points)	Computer Screen (inches)	Printed Maps (inches)	Computer Screen (millimeters)	Printed Maps (millimeters)
1.5	8	6	0.11	0.08	2.8	2.1
2	11	8	0.15	0.11	3.8	2.8
3	16	12	0.22	0.17	5.6	4.3
4	21	16	0.30	0.22	7.5	5.7
5	27	20	0.37	0.28	9.4	7.1
10	53	40	0.74	0.56	18.8	14.2
20	107	80	1.48	1.12	37.6	28.4
30	160	121	2.22	1.68	56.4	42.6
50	266	201	3.70	2.79	94.0	70.9
100	533	402	7.40	5.59	188.0	141.9

https://www.esri.com/arcgis-blog/products/product/mapping/guidelines-for-minimum-size-for-text-and-symbols-on-maps

Legibility (易讀性)



Visual Contrast (視覺對比)

- No graphic factor is as important as contrast.
- The way a sign contrast with its background and adjacent signs determines its visibility.
- Do not assume that maximum contrast is automatically desirable.
- If you vary two signs in two ways (e.g. in both shape and size), the contrast between them will be greater than if you vary only one element.

Visual Contrast (視覺對比)

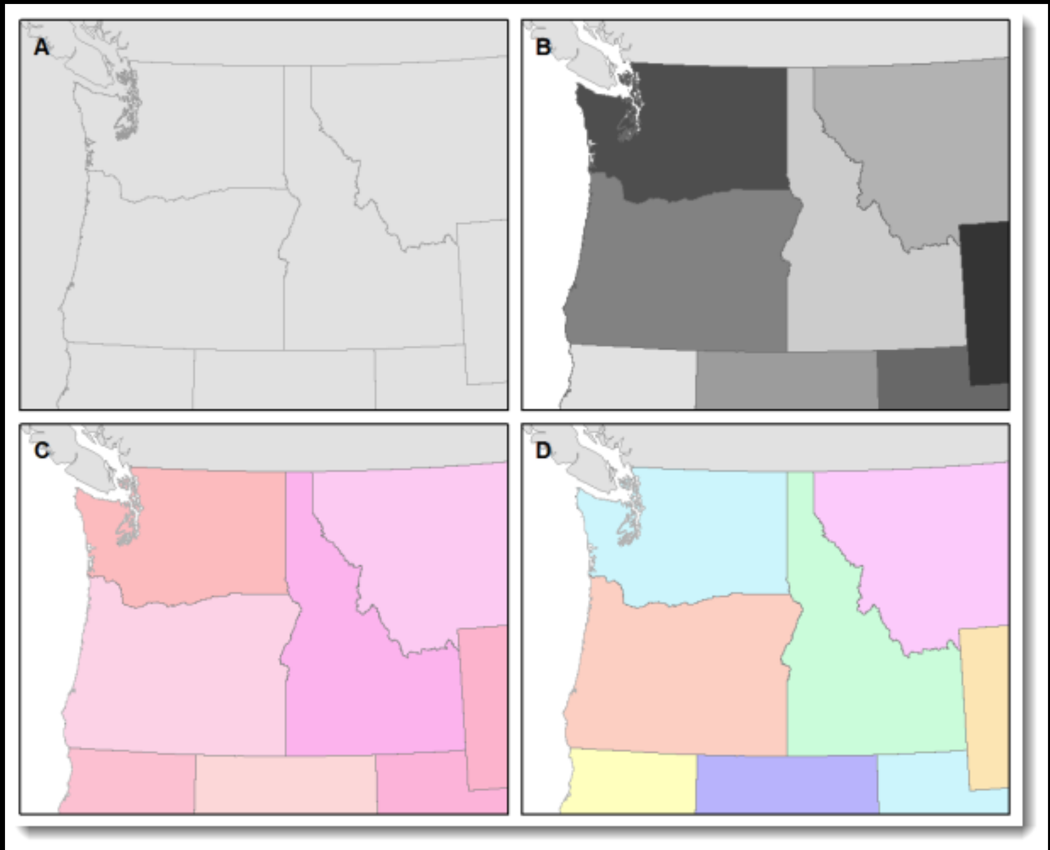


Figure-Ground Organization (圖地組織)

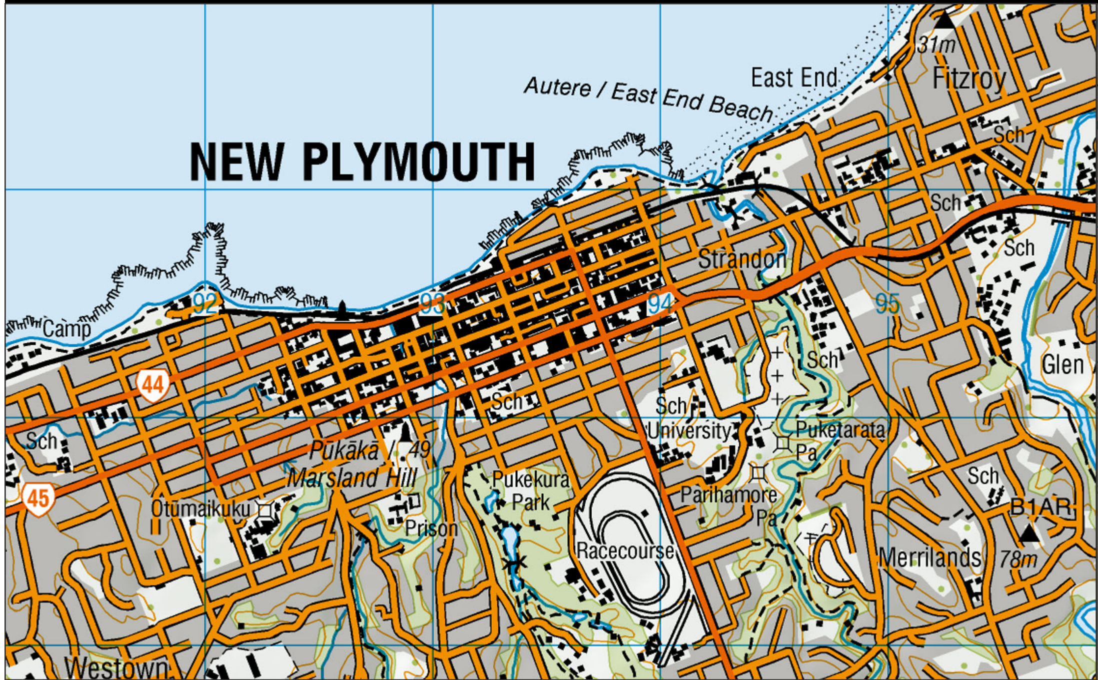
- People immediately organize the display into two contrasting perceptual impressions: a **figure** on which your eye settles, and the amorphous **ground** around it.
- You perceive the figure as a coherent shape or form, with clear outliners in front of or above its surroundings.
- Certain visual stimuli help map readers perceive figure and ground:
  - Differentiation must be present in order for one area to emerge as the figure.
  - Closed forms (封閉形狀) like islands and countries are more likely to be seen as figures.
  - Familiarity (熟悉度) has a great influence on the promotion of figure.
  - Value (明度) difference promotes the emergence of figures.
  - Good contour (良好輪廓) is the graphic equivalent of the term “logical” or ”unambiguous.”
  - Detail (細節) promotes figure.
  - Size (大小) is important in differentiating between figure and ground.

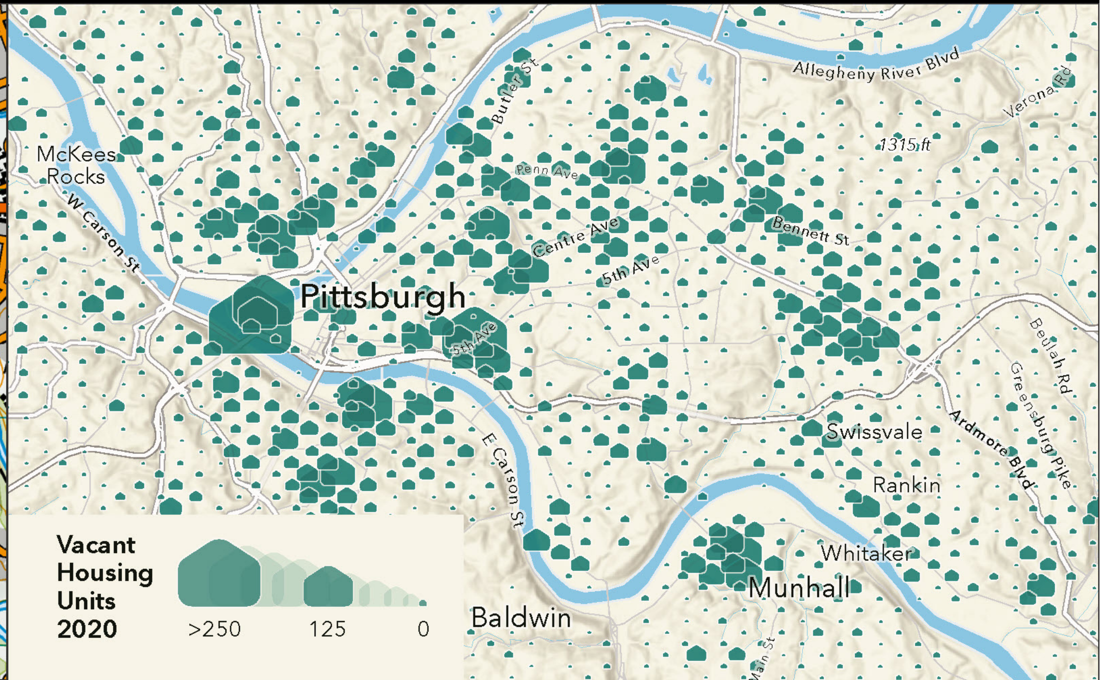
Figure-Ground Organization (圖地組織)

Hierarchical Organization (分層組織)

- Hierarchical organization is the process to set up levels of relative importance.
- Successful map communication requires some kind of internal **graphic structuring**.
- Using different combinations of the visual variables to achieve a **visual layering**.
  - Stereogrammic (立體式) hierarchical organization
  - Extensional (擴充式) hierarchical organization
  - Subdivisional (細分式) hierarchical organization

Hierarchical Organization (分層組織)





Controls on Map Design

- Purpose
- Reality
- Available data
- Map scale
- Audience
- Conditions of use
- Technical limits

Planning of Map Design

- The graphic outline
- Composition
- Visual balance
- Importance of contextual Items
- Orientation and scale

The Graphic Outline

- The outline for a graphic presentation should be graphic.
- Thematic maps usually involve fewer categories of data (than general reference maps), but they pose a different kind of graphic problem.

Composition (Map Components)

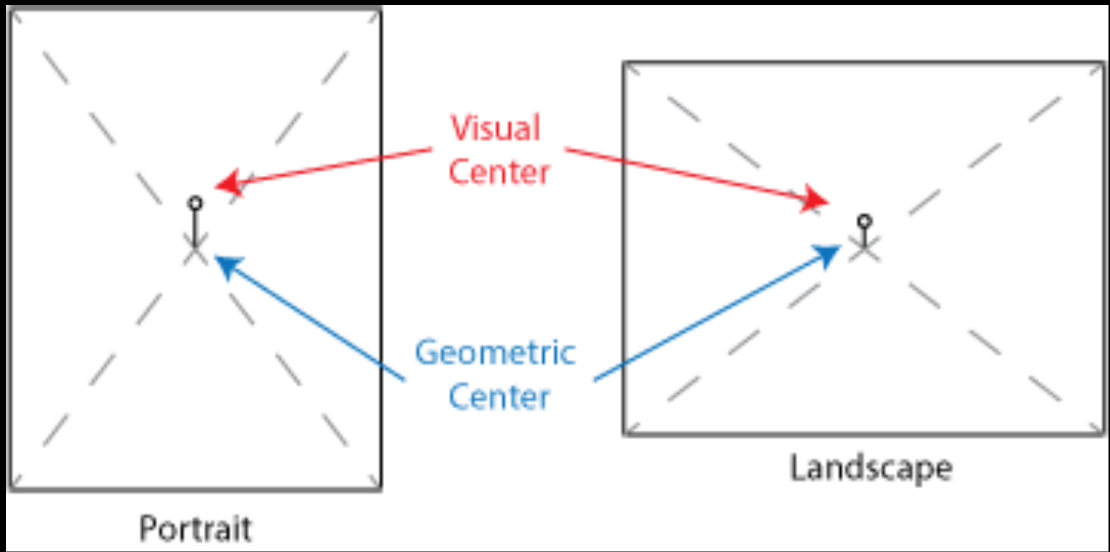
- Standard map components are to identify the place, subject matter, symbolization, orientation, and so on.
- On large-scale series maps, the contextual items usually appear as marginal notations.
- On smaller-scale maps, it is usually convenient to place explanatory items such as titles, legends, and insets within the map frame itself.

Visual Balance

- Balance in graphic design is the positioning of visual components so that their relationship appears logical.
- **Layout** is the process of arriving at proper balance.
- Visual balance depends on the relative position and visual importance of the basic part of a map.
- Visual balance also depends on each item's relation to other items and to the map's **visual center**.
- The format (the size and shape of the image area on which a small-scale map is to appear) is an important factor in balance and layout. Generally, a rectangle with sides having a proportion of ~3:5 seems to be the most pleasing format.

Visual Balance

- The visual center is a point slightly (about five percent of the height) above the center of the bounding shape or the map border.



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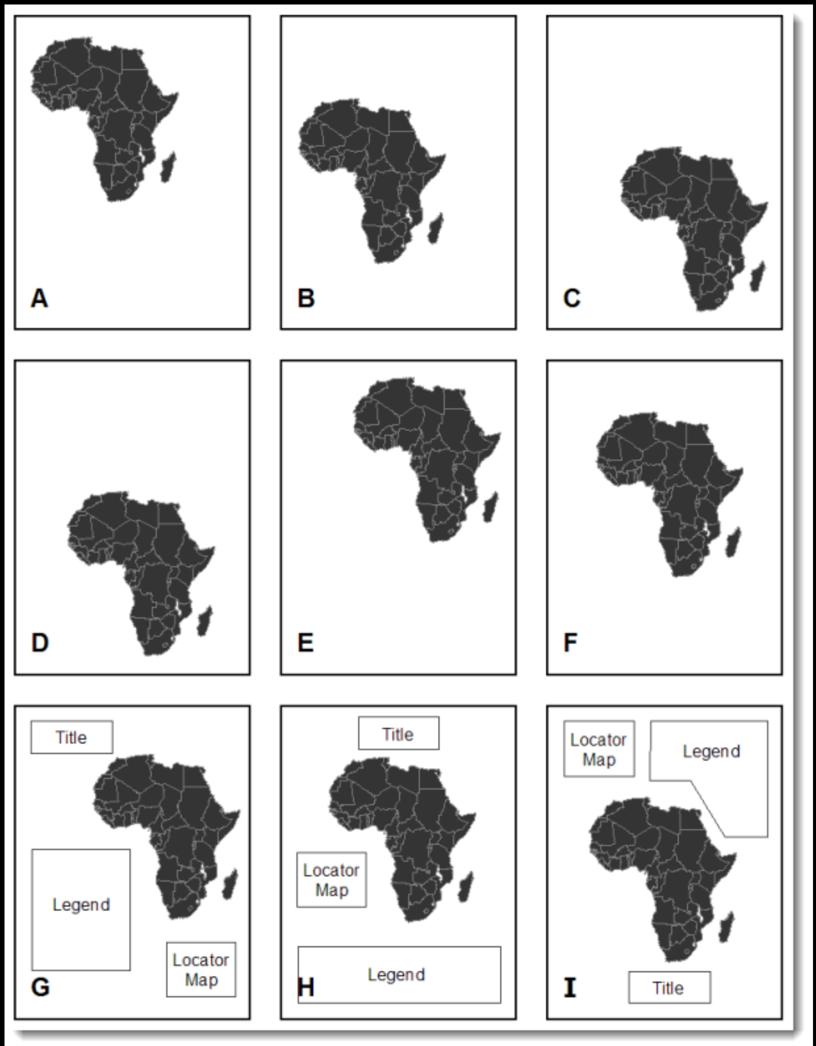
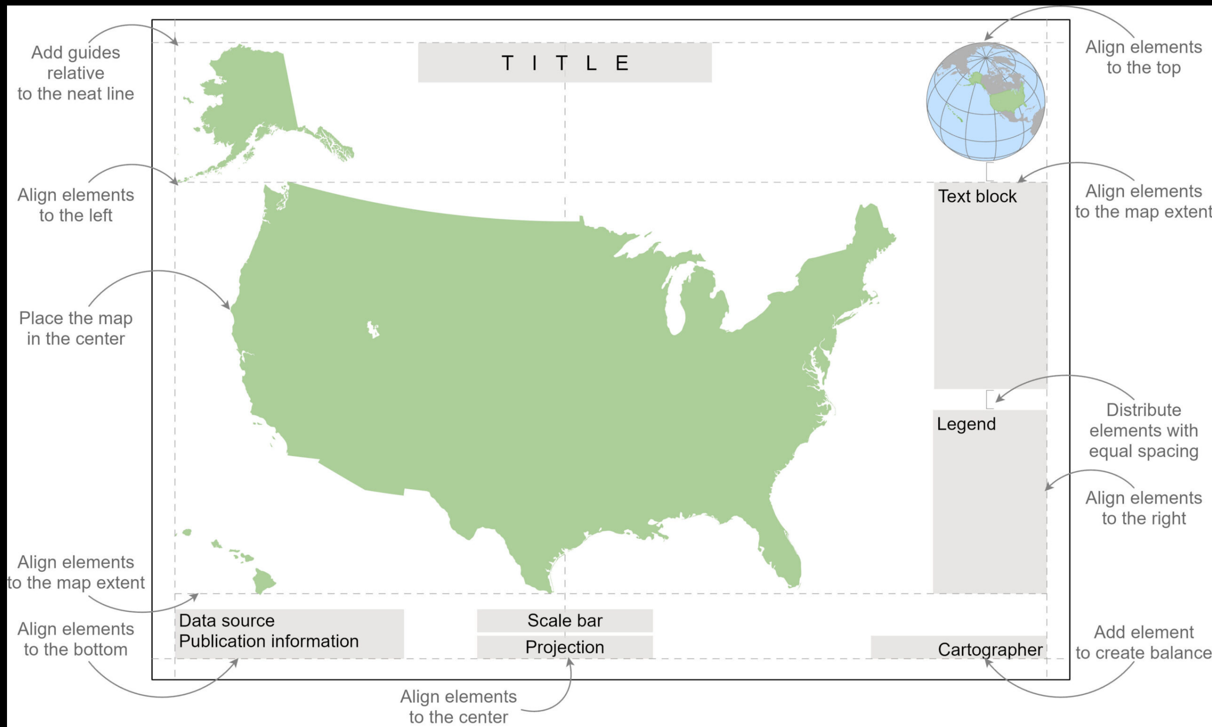
Ancient Islamic World Maps

World Map, 1154, al-Idrisi

Visual Balance





Importance of contextual Items

- You may be inclined to solve the visual balance problem by leaving some map components off the map entirely.
- This is usually not a good idea, and you should carefully consider the importance of each item you are thinking of dropping.
  - Title
    - Sometimes it reveals the map's subject of the area covered by the map. Sometimes a map's subject or area is obvious, and no title is really needed. At other times, the title may be most useful as a shape to help balance composition.
  - Legends
    - In theory, any map symbol that is not self-explanatory should be explained in a legend. In practice, we always take a great deal for granted when creating a map legend, in order to limit legend items to those critical to reading the map in question.
  - Insets
    - When general geographic context of a mapped region is not likely to be obvious to most map users, it is good practice to include one or more inset maps of broader scope. An inset scale is not needed if a global perspective is provided.

Orientation and Scale

- The statement of map orientation and scale also varies in importance from map to map.
- As a rule, if the map has a conventional north-up orientation, a direction indicator is not necessary.
- A direction indicator also map not be needed on global maps where the orientation of familiar geographic feature is obvious, or on maps where a terrestrial reference system is provided as base information.
- A scale indicator is an important factor in making the map useful.
  - Scales should be displayed prominently for line symbols involving distance concepts.
  - For larger-scale maps, the representative fraction (RF) is useful.
  - A graphic scale is much more common on small-scale maps.